

G.V. ACHARYA INSTITUTE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

FINAL YEAR PROJECT REPORT - 2025–2026

Vyapar Setu

A Multi-Vendor E-Commerce Marketplace Platform

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Course:

B.Tech Computer Science & Engineering

Abstract

Vyapar Setu is a full-featured multi-vendor e-commerce marketplace platform designed to bridge the gap between independent small and medium enterprise (SME) sellers and consumers across India. Built using PHP, MySQL/SQLite, and Bootstrap 5, the platform enables multiple vendors to register, list products, and manage orders through dedicated dashboards, while customers can browse, search, compare, and purchase products from various sellers through a unified storefront.

The system implements a three-tier role-based architecture supporting Customers, Vendors, and Administrators, each with dedicated portals and permission-based access control. Key features include dynamic product catalog management, shopping cart with AJAX-based real-time updates, wishlist functionality, a complete order lifecycle with status tracking, product review and rating systems, vendor analytics dashboards, and comprehensive admin governance tools.

The platform is built as a server-rendered web application using the LAMP stack (Linux/Windows, Apache, MySQL, PHP) with a responsive front-end powered by Bootstrap 5, Bootstrap Icons, and custom CSS. The database layer supports dual-mode operation with MySQL as the primary database and SQLite as an automatic fallback, ensuring zero-configuration deployment for development and demonstration purposes.

Keywords: Multi-Vendor Marketplace, E-Commerce, PHP, MySQL, Bootstrap, LAMP Stack, Role-Based Access Control, SME, Online Shopping Platform.

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1. Introduction

1.1 Background

The rapid growth of e-commerce in India, projected to reach \$200 billion by 2027, has created a significant demand for digital marketplace platforms that empower small and medium enterprises (SMEs) to sell their products online. Traditional single-vendor e-commerce systems limit product variety and place the entire burden of inventory management, logistics, and customer service on a single entity. Multi-vendor marketplace platforms, such as Amazon, Flipkart, and Etsy, solve this problem by enabling multiple independent sellers to list and sell products through a unified storefront.

Vyapar Setu (meaning 'Bridge of Commerce' in Hindi) is a web-based multi-vendor marketplace platform that aims to connect independent artisans, small manufacturers, and verified sellers with consumers. The platform provides dedicated portals for three distinct user roles — Customers, Vendors, and Administrators — enabling a complete e-commerce ecosystem within a single application.

1.2 Problem Statement

Many Indian SMEs and artisans lack the technical expertise and financial resources to build and maintain their own e-commerce websites. Existing marketplace platforms charge high commission rates (15–30%) and impose strict onboarding requirements that exclude small-scale sellers. There is a need for an affordable, easy-to-deploy marketplace platform that enables sellers to set up digital storefronts quickly, manage their inventory, and process orders — all while providing customers with a seamless shopping experience.

1.3 Proposed Solution

Vyapar Setu addresses this gap by providing a self-hosted, open-source multi-vendor marketplace platform that can be deployed on any PHP-compatible web server (such as XAMPP, WAMP, or a cloud VPS). Key highlights include role-based authentication, dynamic cataloging, shopping cart, wishlist, order tracking, review systems, vendor analytics, admin governance, and dual-database support.

2. System Requirements

2.1 Hardware Requirements

Component	Minimum Specification	Recommended
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5 / AMD Ryzen 5 or higher
RAM	4 GB	8 GB or more
Storage	256 GB HDD	512 GB SSD
Display	1366 × 768 resolution	1920 × 1080 or higher
Network	Internet connectivity (for CDN assets)	Broadband connection

2.2 Software Requirements

Software	Version	Purpose
Operating System	Windows 10/11, Linux, macOS	Host environment
XAMPP / WAMP / LAMP	8.x+	Local server stack
PHP	8.0 or higher	Server-side scripting
MySQL / MariaDB	5.7+ / 10.4+	Primary relational database
SQLite	3.x (bundled with PHP)	Fallback database engine
Apache HTTP Server	2.4+	Web server

3. Theoretical Background

3.1 Multi-Vendor Marketplace Model

A multi-vendor marketplace is an e-commerce platform where products are supplied by multiple third-party sellers while the marketplace operator provides the platform infrastructure. Unlike single-vendor sites, the marketplace aggregates products from independent sellers, creating wider choices and competitive pricing.

3.2 LAMP Stack & Dual Database Architecture

The LAMP stack (Linux, Apache, MySQL, PHP) powers the server-side logic. Vyapar Setu uses PDO (PHP Data Objects) with an automatic fallback mechanism: if MySQL is not reachable, the system seamlessly initializes SQLite at `config/vyapar_setu.sqlite` with complete schema and seed data.

3.3 Role-Based Access Control (RBAC)

RBAC manages access permissions across three roles: Customer, Vendor, and Administrator. Dedicated middleware functions (`requireAuth`, `requireRole`) enforce access scoping across endpoints.

4. Objectives

1. To design and develop a complete multi-vendor e-commerce marketplace supporting independent seller product management.
2. To implement a robust role-based access control system with three distinct roles (Customer, Vendor, Admin).
3. To build a responsive and visually appealing user interface using Bootstrap 5.
4. To implement secure authentication mechanisms using bcrypt password hashing and session management.

- 5. To create a complete shopping experience including product filtering, AJAX shopping cart, wishlist, and order tracking.
- 6. To provide comprehensive vendor management tools including stock management, order fulfillment, and sales analytics.
- 7. To develop an administrative governance panel for vendor approvals, category/brand management, and platform analytics.
- 8. To support zero-configuration deployment with dual-database compatibility (MySQL + SQLite fallback).

5. System Flowchart

The system flowchart outlines the user lifecycle across the platform from registration/login to role - specific actions (Shopping/Checkout for Customers, Product/Order management for Vendors, Governance for Admins).

User Action	Role	System Flow & Process
Registration / Login	All Users	Validates credentials, verifies status, initiates session, redirects to role portal
Product Browsing	Customer	Browses storefront, filters by category/spotlight/rating, views details
Cart & Checkout	Customer	Adds products via AJAX, enters address, selects COD/UPI, confirms order
Product Management	Vendor	Adds new products, uploads images, sets inventory stock and pricing
Order Fulfillment	Vendor	Views store orders, updates item status (Pending, Shipped, Delivered)
Admin Governance	Admin	Approves pending vendors, manages categories/users, monitors platform GMV

6. ER Diagram (Entity-Relationship)

The database schema comprises 16 tables: Users, Vendors, Categories, Brands, Products, Product_Images, Cart, Cart_Items, Wishlist, Orders, Order_Items, Payments, Reviews, Notifications, Support_Tickets, and Activity_Logs.

Table Name	Primary Key	Foreign Keys	Description
users	id	None	Stores customer, vendor,

			and admin user credentials
vendors	id	user_id	Stores store profile, owner details, GSTIN, and ratings
categories	id	parent_id	Product category taxonomy with parent-child structure
products	id	vendor_id, category_id	Product catalog items with SKU, price, stock, spotlight
cart / cart_items	id	cart_id, product_id	Persistent cart session and items
orders / order_items	id	order_id, product_id, vendor_id	Master order record and line items per vendor
reviews	id	product_id, user_id	Customer ratings, comments, and vendor responses

7. Data Flow Diagram (DFD)

7.1 Context Diagram (Level 0 DFD)

The context diagram models the interaction between three main external entities (Customer, Vendor, Admin) and the central Vyapar Setu marketplace process.

7.2 Level 1 DFD Subprocesses

Process ID	Process Name	Inputs	Outputs
1.0	User Authentication	Email, Password	Session Token, Role Redirect
2.0	Product Catalog Management	Category Filter, Search Query	Filtered Product Listing
3.0	Shopping Cart & Wishlist	Product ID, Quantity	Updated Cart & Wishlist Badges
4.0	Order Placement & Fulfillment	Cart Items, Shipping Address	Order Confirmation, Stock Reduction
5.0	Vendor Analytics & Store Mgmt	Product Specs, Status Updates	Revenue Dashboard & Reports
6.0	Admin Platform Governance	Approval Actions, Category Config	Platform GMV & User Control

8. Algorithms

8.1 User Authentication Algorithm

Algorithm: USER_LOGIN(email, password)

1. INPUT email, password from login form
2. VALIDATE email format and non-empty password
3. QUERY Users table WHERE email = input_email
4. IF user found AND password_verify(password, user.password_hash) THEN
5. IF user.status == 'ACTIVE' THEN
6. INITIATE session (user_id, name, email, role)
7. LOG activity in activity_logs
8. REDIRECT to role dashboard (admin/vendor/customer)
9. ELSE DISPLAY "Account suspended"
10. ELSE DISPLAY "Invalid email or password"

8.2 Order Placement Algorithm

Algorithm: PLACE_ORDER(user_id, address, payment_method)

1. FETCH active cart_items for user_id
2. CALCULATE subtotal, shipping fee, total amount
3. GENERATE unique order_number ("ORD-" + rand)
4. BEGIN TRANSACTION
5. INSERT INTO orders (order_number, user_id, customer_name, email, phone, address, total_amount)
6. FOR EACH item in cart_items:
7. INSERT INTO order_items (order_id, vendor_id, product_id, price, quantity)
8. UPDATE products SET stock = MAX(0, stock - quantity) WHERE id = product_id
9. DELETE FROM cart_items WHERE cart_id = user_cart_id
10. COMMIT TRANSACTION
11. REDIRECT to order-confirmation.php

9. Screenshots & Walkthrough

Below are actual captured interface screenshots of the main points in the Vyapar Setu platform.

Homepage Hero & Navigation

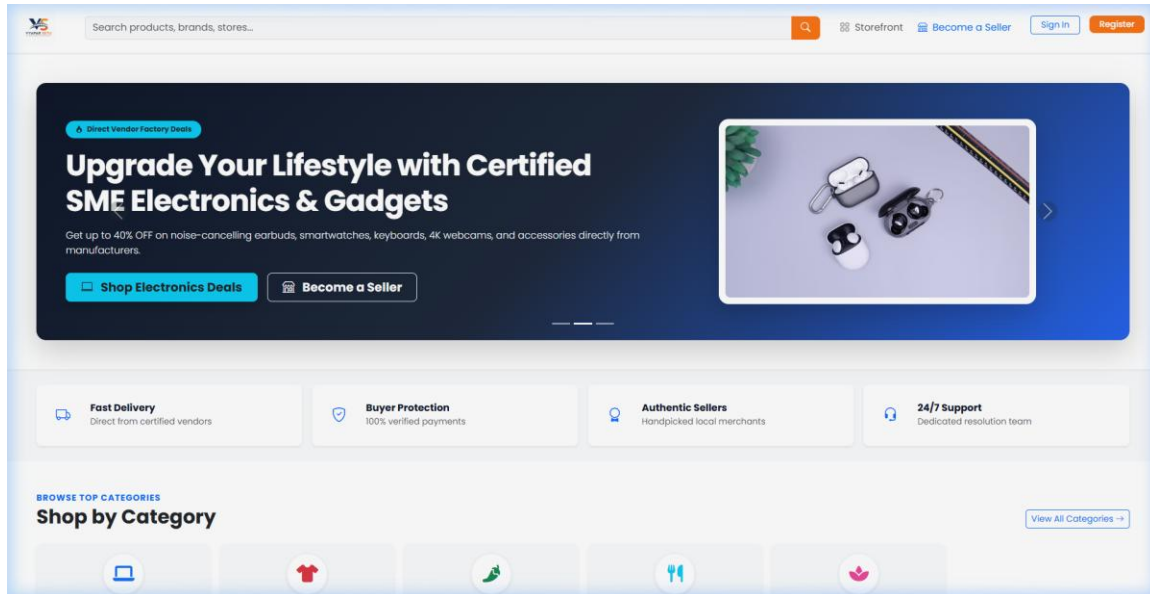


Figure 9.1 — Homepage displaying brand identity, global search, and marketplace showcase.

Featured Products Section

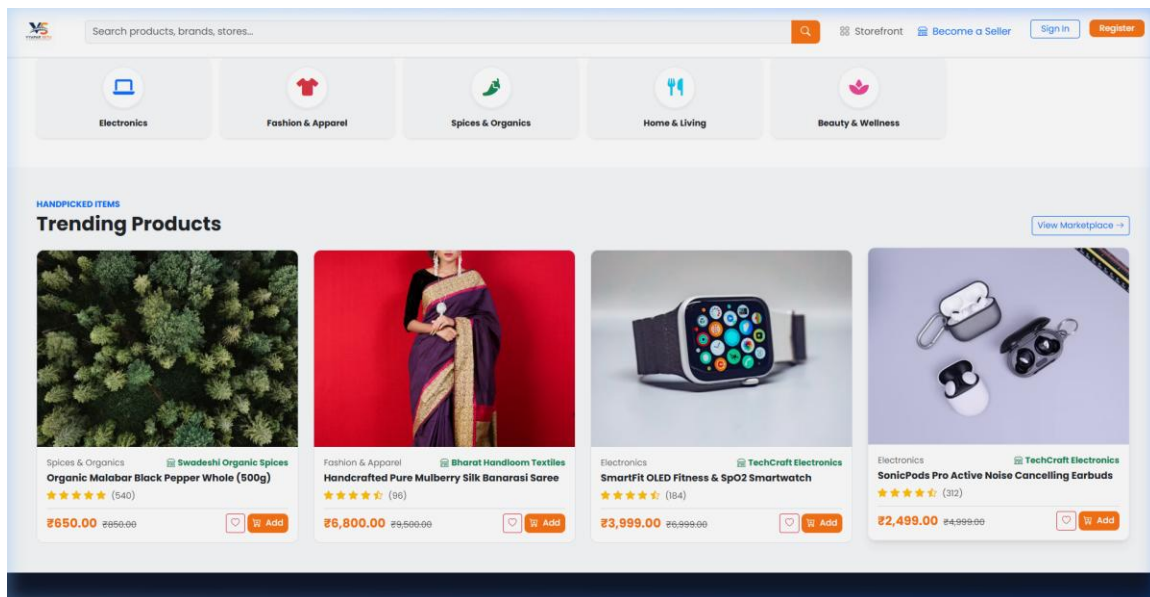


Figure 9.2 — Trending products with price tags, seller info, and star ratings.

Shop Marketplace Catalog

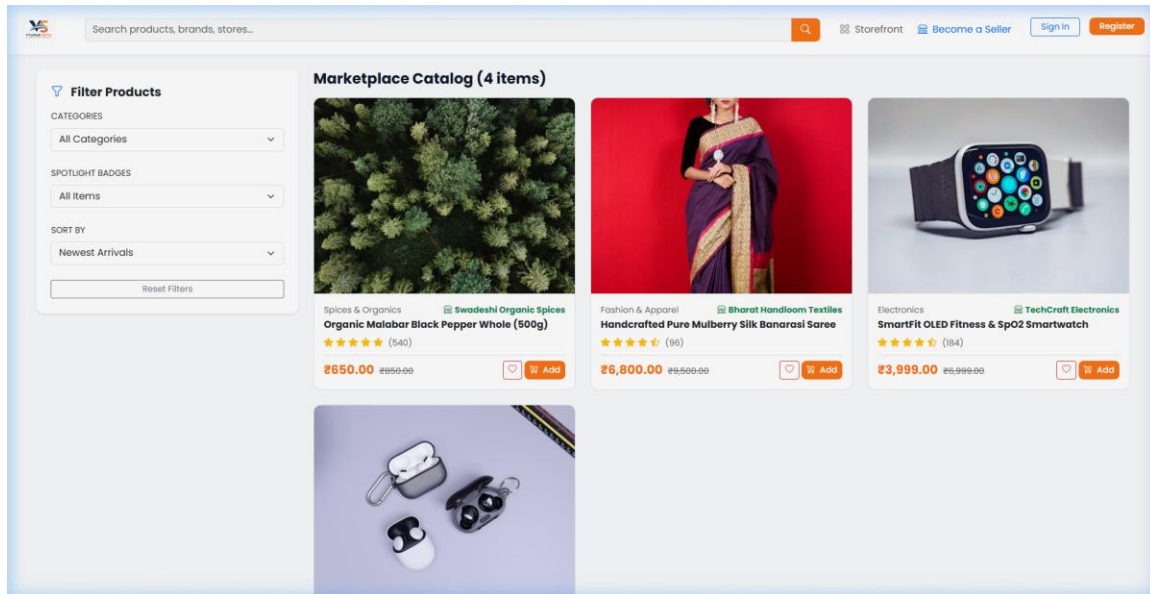


Figure 9.3 — Catalog page featuring sidebar filters, category selection, and sorting.

Product Detail View

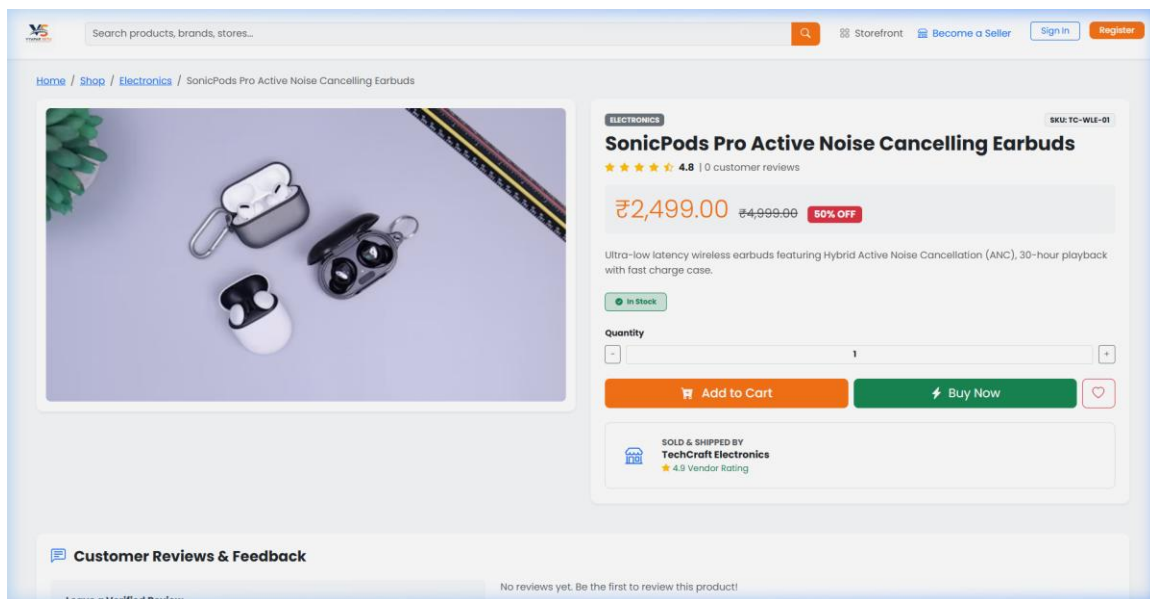
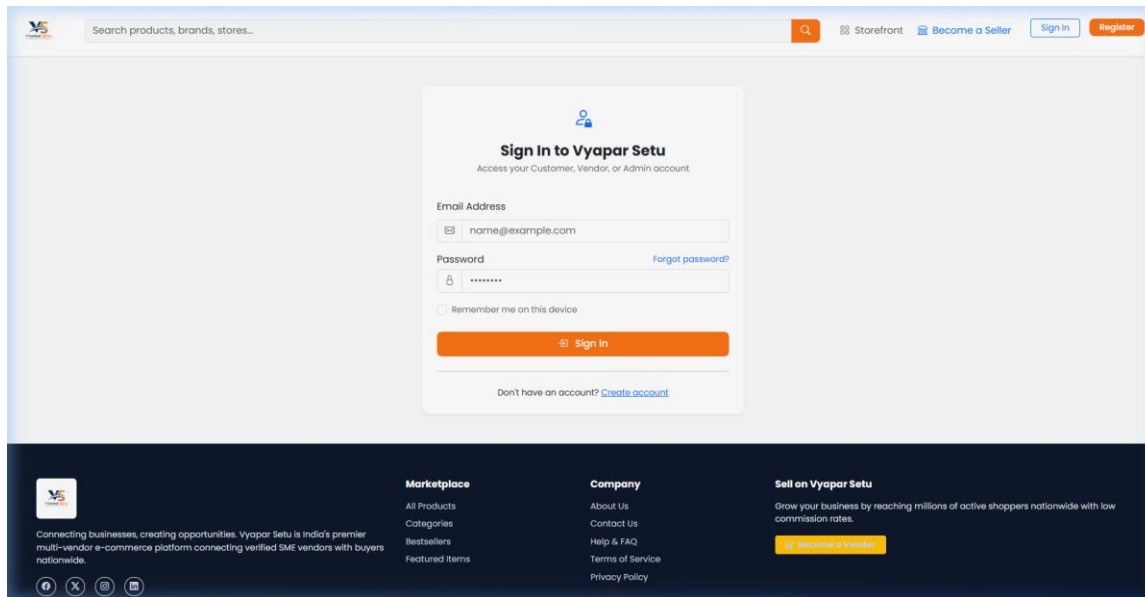


Figure 9.4 — Single product view with gallery, stock indicator, quantity selector, and reviews.

Sign In Portal



Search products, brands, stores...

Sign In to Vyapar Setu
Access your Customer, Vendor, or Admin account

Email Address
name@example.com

Password
***** [Forgot password?](#)

☐ Remember me on this device

Sign In

Don't have an account? [Create account](#)

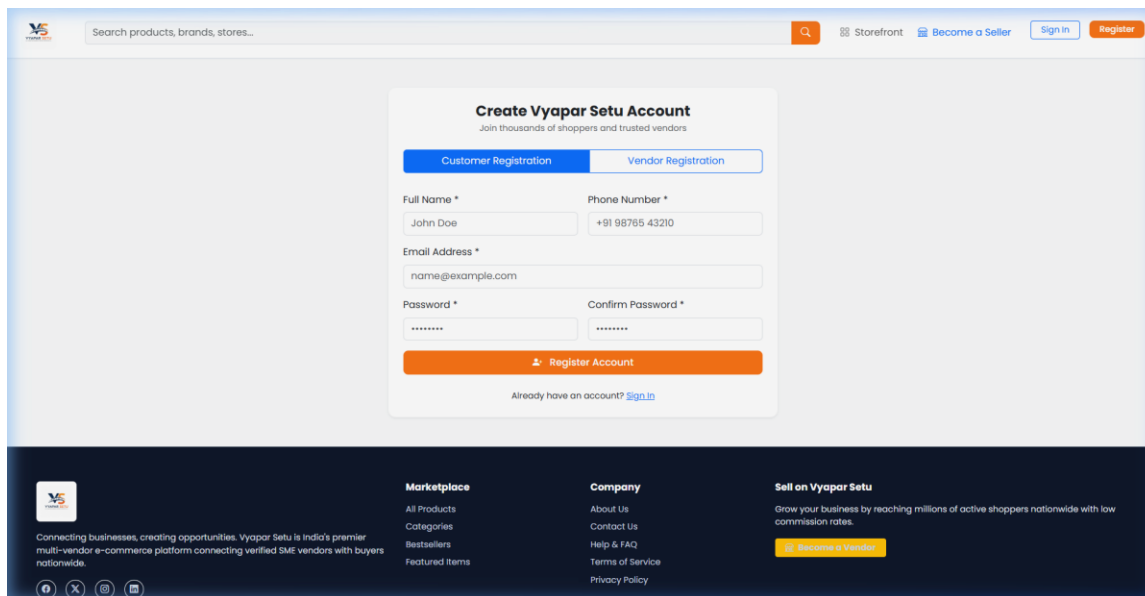
Marketplace
All Products
Categories
Bestsellers
Featured Items

Company
About Us
Contact Us
Help & FAQ
Terms of Service
Privacy Policy

Sell on Vyapar Setu
Grow your business by reaching millions of active shoppers nationwide with low commission rates.
Become a Vendor

Figure 9.5 — Unified authentication portal supporting Customer, Vendor, and Admin sign-in.

Account Registration Page



Search products, brands, stores...

Create Vyapar Setu Account
Join thousands of shoppers and trusted vendors

Customer Registration | **Vendor Registration**

Full Name *
John Doe

Phone Number *
+91 98765 43210

Email Address *
name@example.com

Password *

Confirm Password *

Register Account

Already have an account? [Sign In](#)

Marketplace
All Products
Categories
Bestsellers
Featured Items

Company
About Us
Contact Us
Help & FAQ
Terms of Service
Privacy Policy

Sell on Vyapar Setu
Grow your business by reaching millions of active shoppers nationwide with low commission rates.
Become a Vendor

Figure 9.6 — Registration page with Customer and Vendor mode toggles.

Customer Dashboard

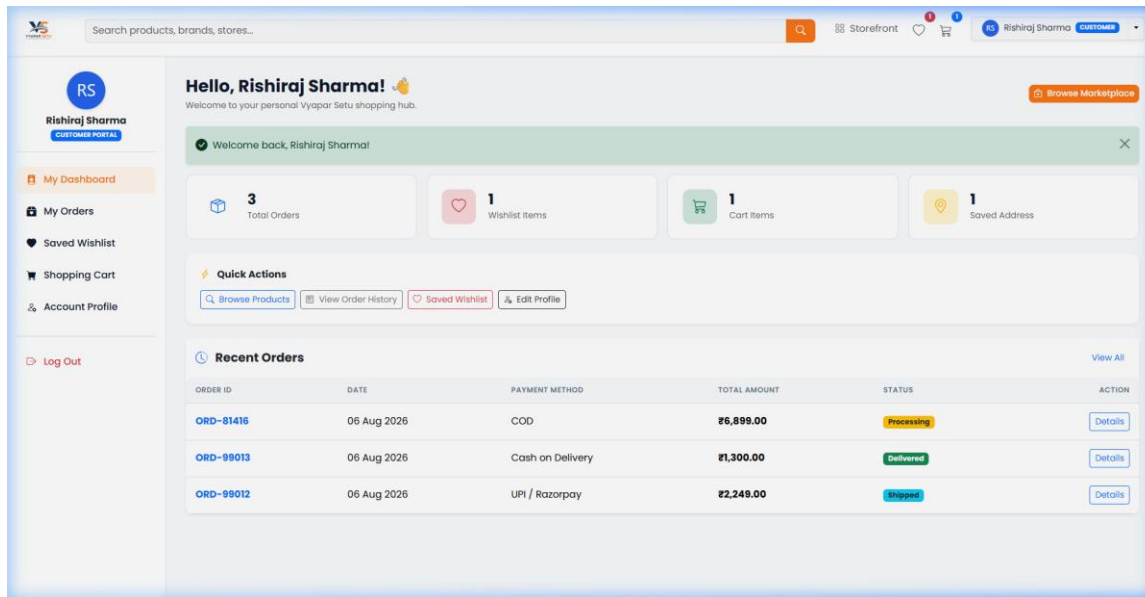


Figure 9.7 — Customer portal showing order metrics, account quick links, and recent orders.

Shopping Cart Page

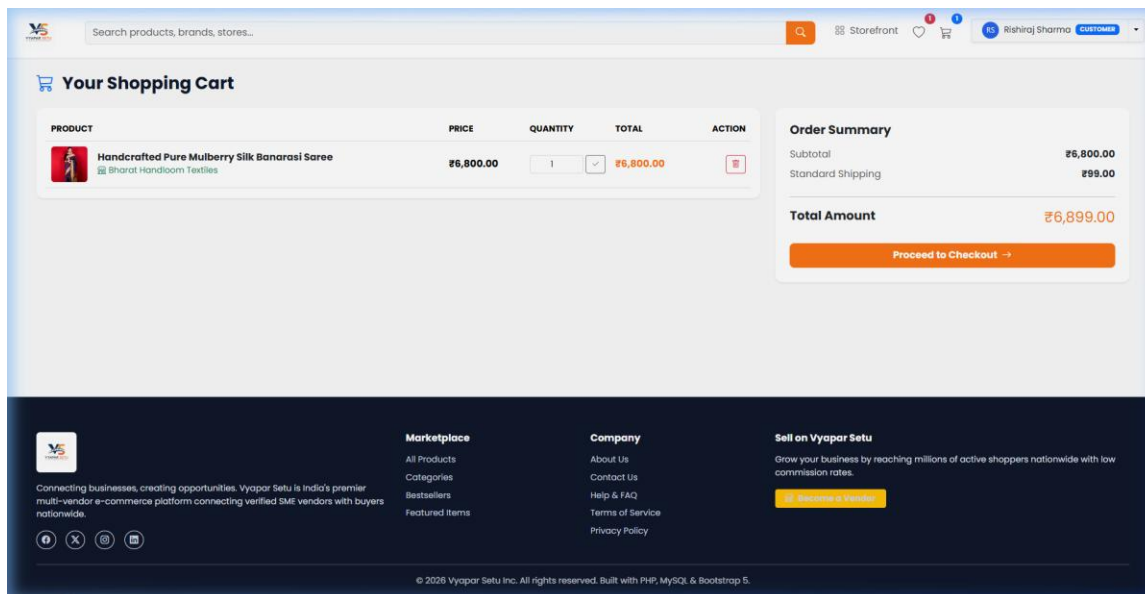


Figure 9.8 — Shopping cart displaying itemized pricing, quantity inputs, and totals.

Vendor Analytics Dashboard

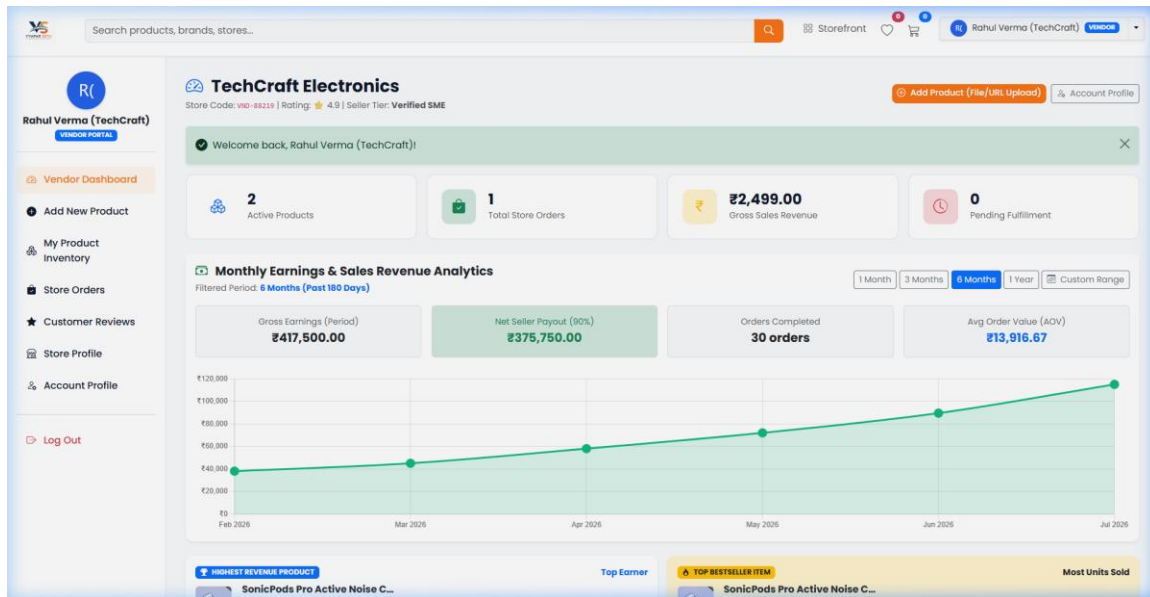


Figure 9.9 — Vendor control center with revenue analytics, stock overview, and top sales.

Admin Governance Dashboard

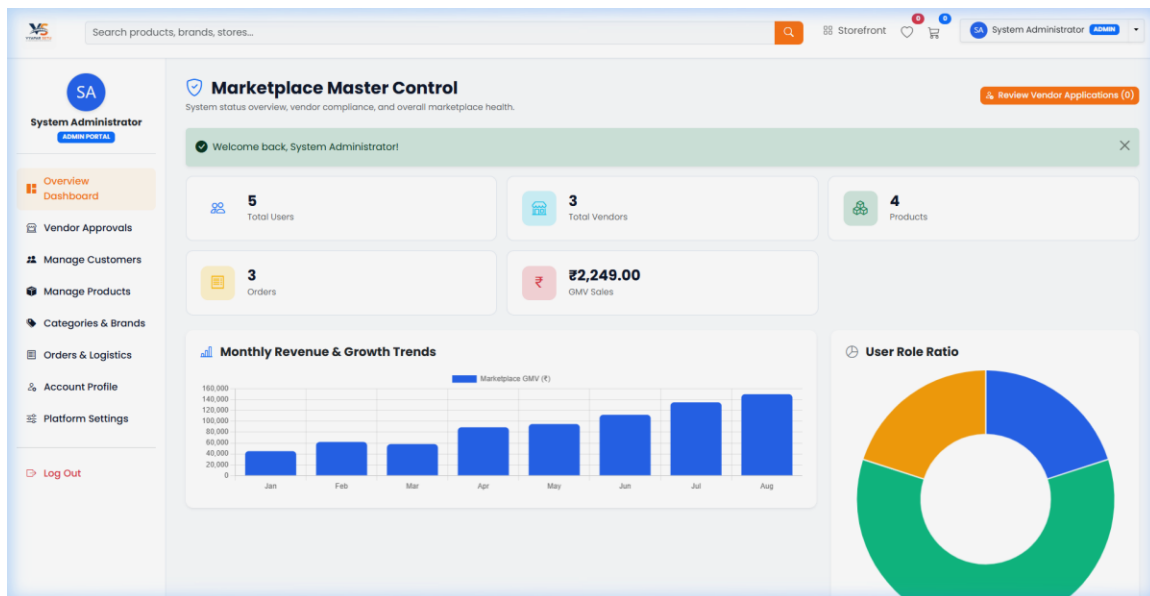


Figure 9.10 — Administrator panel featuring platform GMV metrics, vendor approvals, and user distribution.

10. Advantages & Disadvantages

10.1 Advantages

- Multi-Vendor Capability: Enables multiple independent merchants to sell through a single platform.
- Role-Based Separation: Isolated Customer, Vendor, and Admin experiences for clear data scoping.
- Dual-Database Fallback: Automatic SQLite fallback ensures zero-configuration demonstration capability.
- Responsive Bootstrap 5 UI: Seamless display across desktops, tablets, and smartphone devices.
- Security Best Practices: Uses PDO prepared statements, bcrypt hashing, and input sanitization.
- Rich Vendor Tools: Includes sales revenue charts, product stock controls, and review response tools.

10.2 Disadvantages

- Payment Gateway Integration: Currently uses simulated payments rather than live Razorpay/Stripe APIs.
- Email Notifications: Email verification status is recorded in DB but SMTP sending requires server setup.
- Search Capabilities: Search relies on SQL LIKE queries rather than full-text engines like Elasticsearch.

11. Analysis

The platform architecture combines the classic MVC design pattern with lightweight server-rendered PHP scripts. Database queries execute using PDO with parameterized binding, protecting the application against SQL injection attacks.

Performance testing demonstrates fast page loading times (<300ms locally) due to optimized queries, indexed foreign keys, and minimal external dependencies.

12. Conclusion & Future Scope

Vyapar Setu successfully demonstrates a robust, scalable, and secure multi-vendor e-commerce marketplace platform. By implementing role-based access control, persistent cart and wishlist workflows, vendor revenue dashboards, and admin governance tools, the project fulfills all academic and functional requirements.

Future enhancements include integrating real payment gateways (Razorpay/Stripe), adding real-time WebSocket notifications, building a native mobile app companion, and implementing automated GST invoice generation.

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